RAJALAKSHMI ENGINEERING COLLEGE

Padmapriya S K 251801218-2025-AIDS

P2

Started on	Saturday, 5 September 2026, 11:36 AM
State	Finished
Completed on	Saturday, 5 September 2026, 11:36 AM
Time taken	30 secs
Marks	1.00/1.00
Grade	**10.00** out of 10.00 (**100**%)

Question 1 | Correct Mark 1.00 out of 1.00

Write a Program to Implement the Quick Sort Algorithm

Input Format:
The first line contains the no of elements in the list-n
The next n lines contain the elements.

Output:
Sorted list of elements

For example:

Input	Result
5 67 34 12 98 78	12 34 67 78 98

Answer:

```c
#include <stdio.h>

void quickSort(int arr[], int low, int high) {
    int i, j, pivot, temp;

    if (low < high) {

        pivot = arr[low];
        i = low + 1;
        j = high;

        while (i <= j) {

            while (i <= high && arr[i] <= pivot) {
                i++;
            }

            while (arr[j] > pivot) {
                j--;
            }

            if (i < j) {
                temp = arr[i];
                arr[i] = arr[j];
                arr[j] = temp;
            }
        }


        temp = arr[low];
        arr[low] = arr[j];
        arr[j] = temp;


        quickSort(arr, low, j - 1);
        quickSort(arr, j + 1, high);
    }
}

int main() {
    int n, i;

    scanf("%d", &n);

    int arr[n];

```

```
47
48      for (i = 0; i < n; i++) {
49          scanf("%d", &arr[i]);
50      }
51
52
```

	Input	Expected	Got	
✔	5 67 34 12 98 78	12 34 67 78 98	12 34 67 78 98	✔
✔	10 1 56 78 90 32 56 11 10 90 114	1 10 11 32 56 56 78 90 90 114	1 10 11 32 56 56 78 90 90 114	✔
✔	12 9 8 7 6 5 4 3 2 1 10 11 90	1 2 3 4 5 6 7 8 9 10 11 90	1 2 3 4 5 6 7 8 9 10 11 90	✔

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

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